



Paper ID : 261474

Printed Page: 1 of 1

Subject Code:BP605T

Roll No:

BPHARM

(SEM. VI) THEORY EXAMINATION 2025-26

PHARMACEUTICAL BIOTECHNOLOGY– THEORY

TIME: 3 HRS

M.MARKS: 75

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.**SECTION A**1. Attempt *all* questions in brief.

10x2 = 20

(a)	What is the applications of enzyme immobilization?
(b)	Suggest the roles of lipase and catalase.
(c)	What are the functions of DNA ligase?
(d)	State the mode of action of ELISA
(e)	What is hapten?
(f)	Briefly state the various types of hypersensitivity reactions.
(g)	Mention the storage condition of polio vaccine
(h)	Define monoclonal antibodies
(i)	How will you store BCG Vaccine?
(j)	What is the working principle of a biosensor?

SECTION B2. Attempt any *two* of the following:

2 x 10 = 20

(a)	Elucidate the structure and function of MHC. Explain in details the various types of hypersensitivity reactions.
(b)	Explain in detail Western Blotting . Write down the genetic organization of eukaryotes.
(c)	Suggest the role of recombinant DNA Technology and genetic engineering in the production of hepatitis B Vaccine

SECTION C3. Attempt any *five* part of the following:

7 x 5 = 35

(a)	Narrate the various steps involved in the production of enzyme highlighting the process of enzyme immobilization techniques.
(b)	Describe the processes of transformation, transduction, conjugation and applications of microbial biotransformation.
(c)	Demonstrate the design of large scale production fermenter and outline its various controls.
(d)	Explain with a flow chart the process of production and purification of Penicillin G by the process of fermentation. Suggest the storage requirements of dry human plasma.
(e)	Explain the various types of blood products and plasma substitutes alongwith their specific uses.
(f)	Narrate the working and applications of biosensors in Pharmaceutical Industries.
(g)	Write short notes on microbial biotransformation of steroids